
Is the immobility of advanced dementia a form of lorazepam-responsive catatonia?

Joseph Martin Alisky, MD, PhD

Abstract

Patients with end-stage dementia typically are very immobilized. Could this state actually be a form of lorazepam-responsive catatonia? Catatonia has been documented following cerebrovascular accidents, head injury, HIV encephalitis, brain tumors, and multiple sclerosis. Identified anatomical substrates include frontal lobes, parietal lobes, limbic system, diencephalon, and basal ganglia. Given that Alzheimer's disease, vascular dementia, Lewy body dementia, corticobasal degeneration, frontotemporal dementia, and Parkinsonian dementia often have degeneration in some of the same areas, dramatic awakenings might be possible by giving lorazepam challenges to locked-in dementia patients. If even a small percentage were lorazepam responders, the impact worldwide would be tremendous. Serious consideration should be given to undertaking large-scale clinical trials.

Key words: lorazepam, catatonia, dementia

Case report

A few months ago, I evaluated a patient with advanced Alzheimer's disease and vascular dementia, who had been seen holding his arms in bizarre postures for hours at a time. I recommended administering 2 mg of parenteral lorazepam to see if the patient had catatonia, because catatonic patients can dramatically recover in a short time following intramuscular, intravenous, and oral lorazepam.¹⁻³ The patient's family declined, but the experience raised a question that may interest a broad audience of dementia researchers and other parties. What if many patients with end-stage dementia, in fact, have unrecognized lorazepam-responsive catatonia?

Joseph Martin Alisky, MD, PhD, Marshfield Clinic Research Foundation, Marshfield, Wisconsin; Marshfield Clinic-Thorp Center, Thorp, Wisconsin.

Discussion

The last stages of Alzheimer's disease and other dementias often contain the three cardinal features of catatonia:

1. little or no spontaneous movement;
2. mutism; and
3. refusal to eat or drink.¹

Catatonia often has other features such as waxy flexibility, extreme rigidity of muscles, echolalia, and apraxia.¹ However, these secondary features are not required to make a diagnosis of catatonia if the three cardinal manifestations are present.¹

The notion of catatonia as part of the clinical spectrum of dementia is plausible, if one considers that catatonia has been described as a sequela of many other disparate processes of neuronal injury, including cerebrovascular accidents,^{4,5} head injury,⁶ HIV encephalitis,^{7,8} brain tumors,⁹ and multiple sclerosis.¹⁰ Lesions in the frontal lobe, parietal lobe, limbic system, diencephalon, and basal ganglia can produce catatonia.⁵ Thus, there is an anatomical basis for catatonia with all of the major dementias, including Alzheimer's disease, vascular dementia, Lewy body dementia, corticobasal degeneration, frontotemporal dementia, and Parkinsonian dementia. What would happen if large numbers of locked-in dementia patients were given lorazepam challenges? With conventional catatonia, over 70 percent of patients usually respond to lorazepam, and refractory cases often improve with electroconvulsive therapy (ECT).¹⁻³ If even a small fraction of late-stage dementia patients were lorazepam responders, say 1 to 10 percent, worldwide, hundreds of thousands to millions of patients could go from being *non compos mentis* to being capable of some meaningful interaction. The reduction in cost of

care and the corresponding increase in quality of life would be immense. Furthermore, these same patients might subsequently respond to other therapeutic agents such as cholinesterase inhibitors, atypical antipsychotics, antidepressants, and memantine to regain even more function. With the possibility of huge benefits and almost no risk of side effects (lorazepam is a very benign medication if given under monitored circumstances where hypotension and respiratory depression can be obviated), large-scale clinical trials might be very useful indeed, to potentially roll back the last stages of dementia. *Audaces fortuna juvat* (fortune favors the bold).

Acknowledgments

The author received no funding for the preparation of this manuscript and has no proprietary relationship with the makers of any medications or treatments mentioned herein.

References

1. Rosebush PI, Hildebrand AM, Furlong BG, et al.: Catatonic syndrome in a general psychiatric inpatient population: Frequency, clinical

presentation and response to lorazepam. *J Clin Psychiatry*. 1990; 51(9): 357-362.

2. Bush G, Fink M, Petrides G, et al.: Catatonia. II. Treatment with lorazepam and electroconvulsive therapy. *Acta Psychiatr Scand*. 1996; 93(2): 135, 137-143.

3. Koek RJ, Mervis JR: Treatment-refractory catatonia, ECT, and parenteral lorazepam. *Am J Psychiatry*. 1999; 156(1): 160-161.

4. Howard RJ, Low-Beer TS: Catatonia following biparietal infarction with spontaneous recovery. *Postgrad Med J*. 1989; 65(763): 316-317.

5. Saver JL, Greenstein P, Ronthal M, et al.: Asymmetric catalepsy after right hemisphere stroke. *Mov Disorder*. 1993; 8(1): 69-73.

6. Wilcox JA, Nasrallah HA: Organic factors in catatonia. *Br J Psychiatry*. 1986; 149: 782-784.

7. Snyder S, Prenzlauer S, Maruyama N, et al.: Catatonia in a patient with AIDS-related dementia. *J Clin Psychiatry*. 1992; 53(11): 414.

8. Scamvougeras A, Rosebush PI: AIDS-related psychosis with catatonia responding to low-dose lorazepam. *J Clin Psychiatry*. 1992; 53(11): 414-415.

9. Sheline YI, Miller MB: Catatonia relieved by oral diazepam in a patient with a pituitary microadenoma. *Psychosomatics*. 1986; 27(12): 860-862.

10. Mendez MF: Multiple sclerosis presenting as catatonia. *Int J Psychiatry Med*. 1999; 29(4): 435-441.

Call for Papers

American Journal of Alzheimer's Disease and Other Dementias
invites the submission of dementia-related articles in the following areas:
research, case studies, literature reviews, public policy, opinion and commentary.

Some suggested topics include:

- Evidence-based practice
- Practice-based clinical research
- Laboratory research in neurology and genetics
- Pharmacology
- Nonpharmacological therapies
- Administrative issues

For more information, contact:

Editorial Department
American Journal of Alzheimer's Disease and Other Dementias
470 Boston Post Road
Weston, Massachusetts 02493

Tel: 781-899-2702 • **Fax:** 781-899-4900

E-mail: alzheimers@pnpc.com

Web site: www.pnpc.com